

MAIA

Intelligent Monitoring & Auto-Healing Platform

Executive Overview

Version 2.1 · May 2026

What is MAIA?

MAIA (*Monitoring Assistant & Intelligent Automation*) is a rule-based intelligent automation platform that monitors your data pipelines 24/7, detects failures instantly, classifies them automatically, and either fixes them without human intervention or tells your operators exactly what to do.



Faster Detection

Seconds vs. hours of manual log review



Less Manual Work

Auto-heal rules eliminate repetitive approvals



Full Audit Trail

Every action recorded immutably

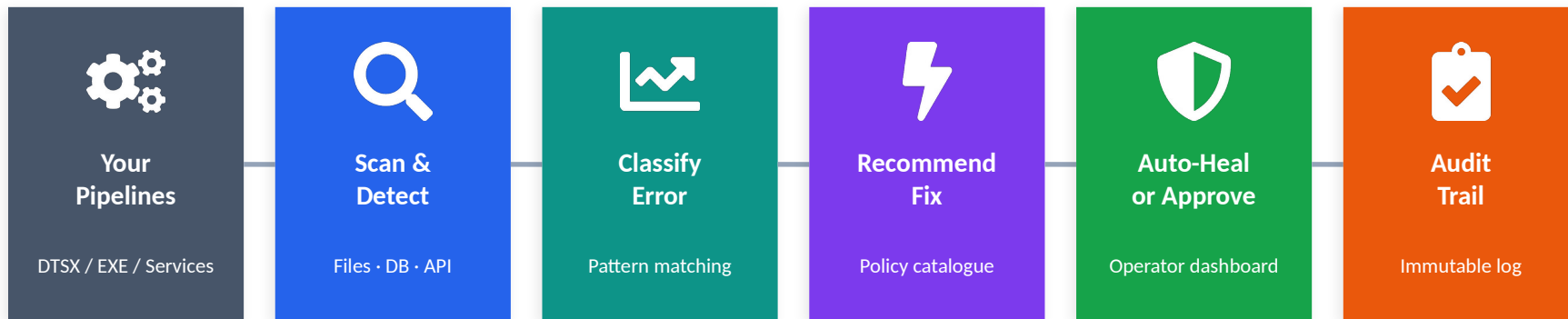


Self-Service Config

No database access needed

How It Works

MAIA watches your pipelines, understands what went wrong, and either fixes it automatically or tells your operator exactly what to do.



Every step is rule-based and deterministic — fully auditable, offline-capable, and predictable.

Platform Capabilities



Multi-Source Monitoring

Scans log files, database tables, and API endpoints. Byte-offset and row-level watermarks prevent re-processing.



Intelligent Classification

Pattern-matching rules classify every failure by error type. Priority-ordered, per-job overrides, wildcard support.



Automated Fix Execution

Configurable fix policies per error type: HTTP calls, stored procedures, SQL scripts, or manual routing.



Auto-Heal

Known issues resolve automatically without operator intervention. Every auto-heal is logged to the audit trail.



Operator Dashboard

Modern web UI for reviewing failures, approving fixes, and managing configuration — no database access required.

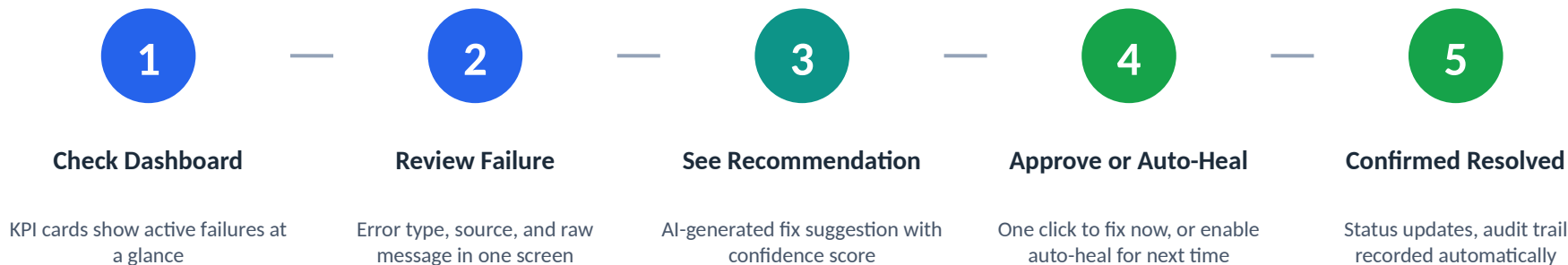


Complete Audit Trail

Immutable AuditLog records every detection, classification, fix execution, and operator decision.

Operator Experience

A typical operator shift takes 5 minutes of review instead of hours of log investigation.



Key Screens

Dashboard — KPI cards & failure overview

Failure Detail — error context & recommendations

Recommendations — batch review with auto-heal toggles

Config — manage jobs, rules & fix policies (no DB access)

Audit Log — immutable history of all actions

Business Impact

Before MAIA

- ✗ Manual log scanning every morning
- ✗ Hours to identify root cause
- ✗ Same fixes repeated daily by operators
- ✗ Fix history tracked in spreadsheets
- ✗ No visibility into overnight failures
- ✗ Dependent on individual operator knowledge

With MAIA

- ✓ Continuous automated monitoring 24/7
- ✓ Failures detected and classified in seconds
- ✓ Known issues auto-healed without intervention
- ✓ Every action recorded in immutable audit log
- ✓ Real-time dashboard with status at a glance
- ✓ Knowledge encoded in configurable rules

90%

Less manual
log review time

24/7

Continuous
monitoring

100%

Audit trail
coverage

<5 min

Operator shift
review time

Roadmap — Current & Next Generation

Today — v2.1 (Built & Operational)

- ✓ Rule-based classification engine
- ✓ Three scan strategies (File, DB, API)
- ✓ Automated fix execution (4 executor types)
- ✓ Auto-heal with configurable policies
- ✓ Full operator dashboard & config UI
- ✓ Immutable audit trail
- ✓ Multi-instance worker (lease-coordinated)
- ✓ Scan history & retention management

Next Generation — Planned

LLM-Assisted Classification

Classify novel errors that don't match existing rules

Dynamic Fix Suggestions

Language model generates fix candidates for review

Anomaly Detection

ML-based detection beyond threshold rules

Role-Based Access Control

Operator, Admin, and Auditor permissions

Architecture is designed for LLM/ML extension — IClassificationStrategy, IFixCatalogue, and IScanStrategy are the natural injection points.

Summary

- MAIA is operational today — all core features built and deployed
- Rule-based automation: predictable, auditable, offline-capable
- Monitors any job type — DTSX, EXE, services, and more
- Auto-heal eliminates repetitive manual work for known issues
- Full audit trail satisfies compliance requirements
- Architecture ready for LLM/ML upgrade when the time is right

MAIA — Monitoring Assistant & Intelligent Automation

Version 2.1 • May 2026 | CONFIDENTIAL — Internal Use Only